

SAP Analytics Cloud
Student Analytics Project

Project Title:

“Educational Challenges You Aim to Address”

Submitted by: Meesala Triveni

Guide: Mr. Raghavendra Rao

Platform: SAP Analytics Cloud (SAC)

Track: Analytics Track

Date: July 27, 2026

Table of Contents

1. Introduction	3
2. Problem Statement	3
3. Objectives	3
4. Tools and Technology Used	3
5. Dataset Description	4
5.1 Dataset in SAC (Data Preparation View)	4
5.2 Dataset Successfully Saved	5
6. Methodology	5
6.1 SAP Analytics Cloud Home Page	6
7. Dashboard / Story Design	6
7.1 Chart Descriptions	7
8. Key Insights and Findings	7
9. Recommendations	8
10. Conclusion	8
11. References	8

1. Introduction

Education is one of the most important pillars of individual and national development, yet several factors continue to affect student performance, attendance, and retention. This project uses SAP Analytics Cloud (SAC) to analyze a dataset of student records and uncover the key educational challenges that impact learning outcomes, such as low attendance, poor academic performance, and student dropout risk.

SAP Analytics Cloud is a cloud-based Business Intelligence (BI) platform that combines analytics, planning, and predictive capabilities in a single environment. It allows users to import data, model it, and build interactive dashboards (Stories) to visualize trends and generate actionable insights. This project demonstrates the end-to-end workflow of SAC — from dataset creation to dashboard-driven insight generation.

2. Problem Statement

Educational institutions often struggle to identify which students are at risk of underperforming or dropping out until it is too late to intervene. Factors such as gender, department, internet access, and family income can significantly influence a student's attendance and academic marks, but these relationships are not always visible without proper data analysis.

This project aims to address the following educational challenge:

“How do factors such as attendance, department, internet access, and family income influence student performance and dropout risk?”

3. Objectives

- To analyze student attendance and marks across different departments.
- To identify the distribution of dropout risk (High, Medium, Low) among students.
- To study the impact of family income on student performance.
- To build interactive dashboards using SAP Analytics Cloud for data-driven decision-making.
- To help institutions identify at-risk students early and plan interventions accordingly.

4. Tools and Technology Used

Platform	SAP Analytics Cloud (SAC)
Modules Used	Datasets, Data Modeler, Stories (Dashboards)
Data Format	Structured tabular dataset (Student records)
Chart Types Used	Bar Chart, Pie Chart, Donut Chart

Dataset Size	100 rows, 8 columns
---------------------	---------------------

5. Dataset Description

The dataset used for this project is titled “Educational_Challenges_SAC_Dataset” and contains 100 student records with 8 columns, consisting of 2 measures and 6 dimensions, as summarized below.

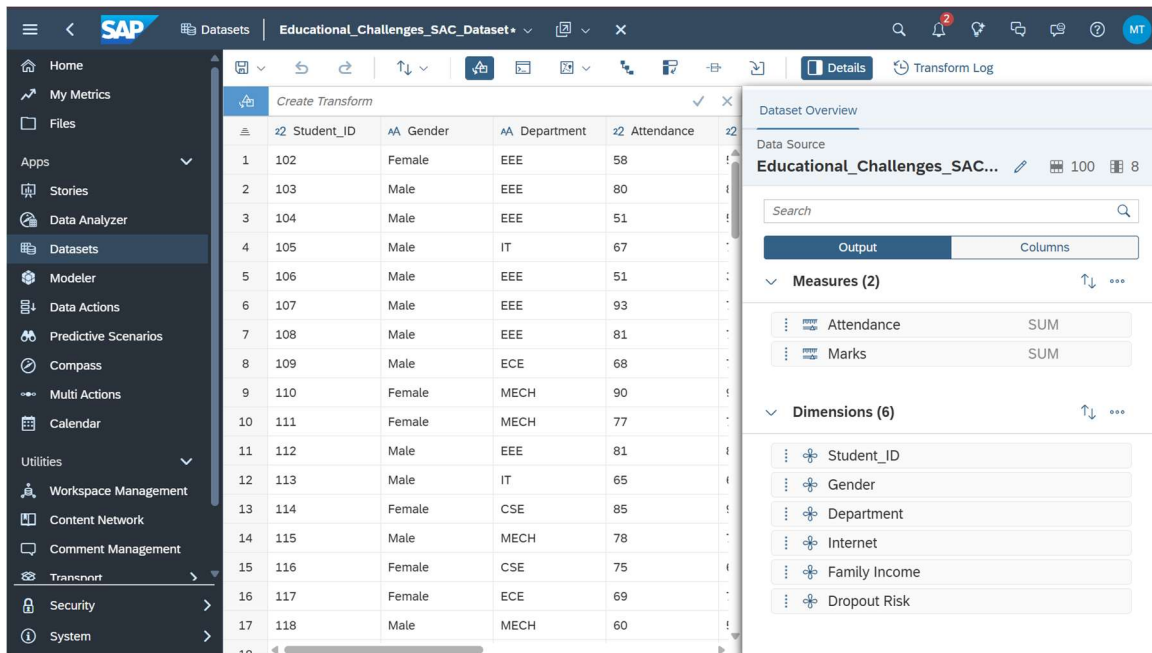
Dataset Name	Educational_Challenges_SAC_Dataset
Number of Rows	100
Number of Columns	8
Measures (2)	Attendance (SUM), Marks (SUM)
Dimensions (6)	Student_ID, Gender, Department, Internet, Family Income, Dropout Risk

Column Details:

- Student_ID – Unique identifier for each student.
- Gender – Male / Female.
- Department – Academic department of the student (CSE, ECE, EEE, IT, MECH).
- Attendance – Percentage attendance of the student.
- Marks – Academic marks scored by the student.
- Internet – Availability of internet access at home.
- Family Income – Income category of the student's family (High, Medium, Low).
- Dropout Risk – Risk level of the student dropping out (High, Medium, Low).

5.1 Dataset in SAC (Data Preparation View)

The screenshot below shows the dataset loaded into SAP Analytics Cloud's Data preparation tool, with the raw student records displayed in the grid and the Dataset Overview panel showing the identified measures and dimensions.

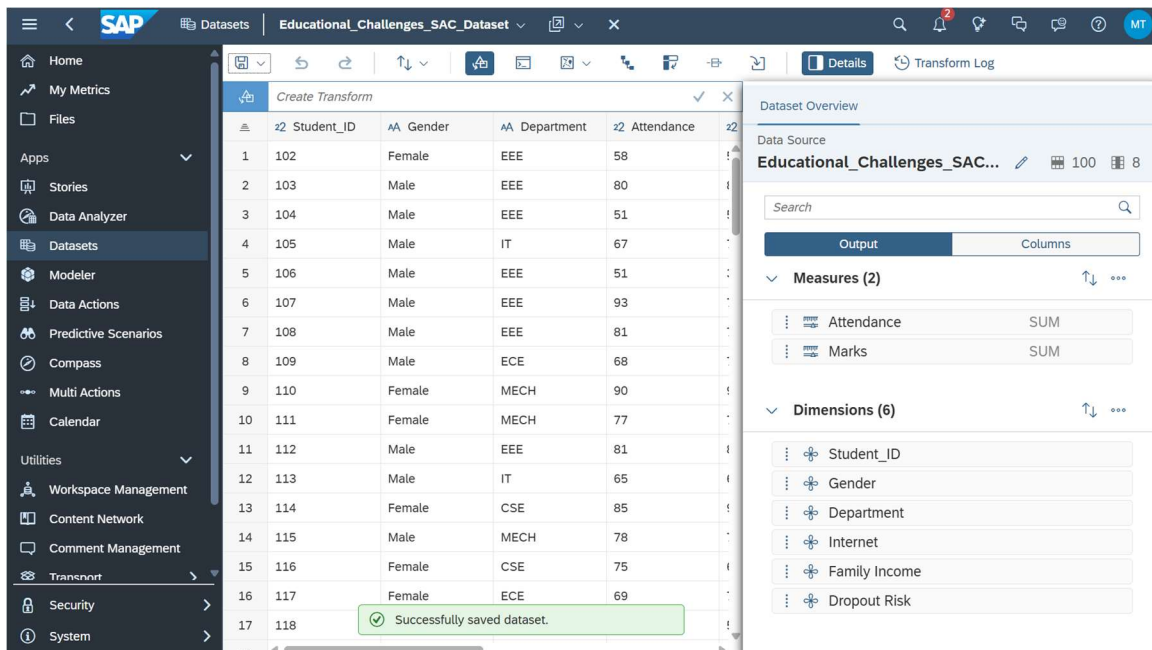


	z2 Student_ID	AA Gender	AA Department	z2 Attendance
1	102	Female	EEE	58
2	103	Male	EEE	80
3	104	Male	EEE	51
4	105	Male	IT	67
5	106	Male	EEE	51
6	107	Male	EEE	93
7	108	Male	EEE	81
8	109	Male	ECE	68
9	110	Female	MECH	90
10	111	Female	MECH	77
11	112	Male	EEE	81
12	113	Male	IT	65
13	114	Female	CSE	85
14	115	Male	MECH	78
15	116	Female	CSE	75
16	117	Female	ECE	69
17	118	Male	MECH	60

Figure 1: Educational_Challenges_SAC_Dataset loaded in the SAC Data preparation editor

5.2 Dataset Successfully Saved

Once the columns were reviewed and validated, the dataset was saved in SAP Analytics Cloud for further modeling and story creation, as confirmed by the success notification shown below.



	z2 Student_ID	AA Gender	AA Department	z2 Attendance
1	102	Female	EEE	58
2	103	Male	EEE	80
3	104	Male	EEE	51
4	105	Male	IT	67
5	106	Male	EEE	51
6	107	Male	EEE	93
7	108	Male	EEE	81
8	109	Male	ECE	68
9	110	Female	MECH	90
10	111	Female	MECH	77
11	112	Male	EEE	81
12	113	Male	IT	65
13	114	Female	CSE	85
14	115	Male	MECH	78
15	116	Female	CSE	75
16	117	Female	ECE	69
17	118	Male	MECH	60

Figure 2: Confirmation message after successfully saving the dataset

6. Methodology

The project was carried out in the following steps using SAP Analytics Cloud:

- Step 1: Logged into SAP Analytics Cloud and accessed the Home page to begin the project.
- Step 2: Created a new dataset named “Educational_Challenges_SAC_Dataset” by importing the student data file.
- Step 3: Used the Data preparation tool to review and validate the 6 dimensions and 2 measures in the dataset.
- Step 4: Saved the dataset for use in dashboard/story creation.
- Step 5: Created a new Story (dashboard) and added multiple charts — bar charts, a pie chart, and a donut chart — to visualize attendance, marks, dropout risk, and family income.
- Step 6: Analyzed the visualizations to derive insights on educational challenges.

6.1 SAP Analytics Cloud Home Page

The project workflow began on the SAP Analytics Cloud Home page, which provides quick access to datasets, stories, and AI-powered “Just Ask” insights.

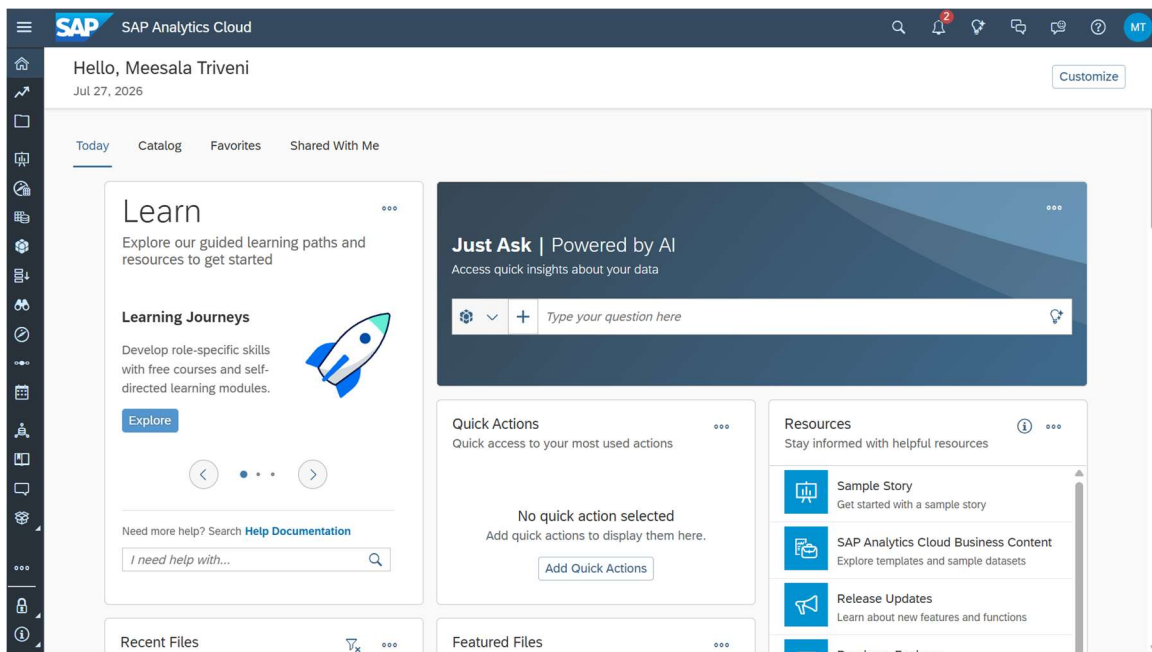


Figure 3: SAP Analytics Cloud Home page (Today tab)

7. Dashboard / Story Design

An interactive Story (dashboard) was built in SAP Analytics Cloud using the prepared dataset. The dashboard consists of four key visualizations that together highlight the educational challenges being addressed by this project.

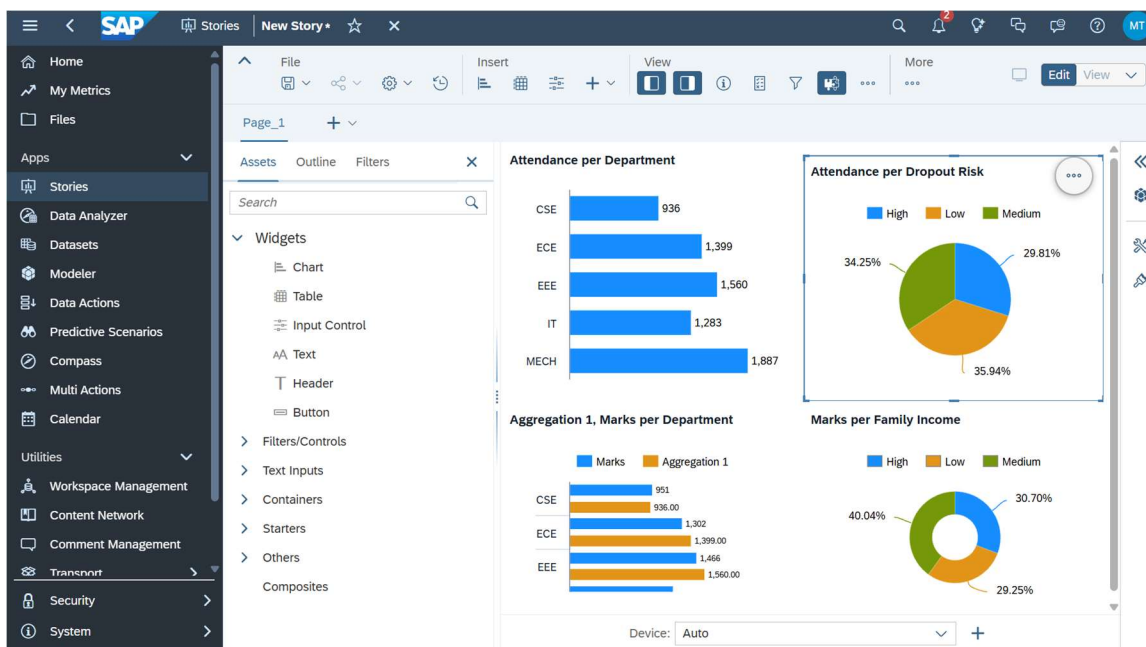


Figure 4: Final Story dashboard — Educational Challenges Analysis

7.1 Chart Descriptions

Attendance per Department (Bar Chart)	Shows total attendance across CSE, ECE, EEE, IT, and MECH departments; MECH (1,887) and EEE (1,560) show the highest cumulative attendance.
Attendance per Dropout Risk (Pie Chart)	Breaks down attendance by dropout risk category — Low (35.94%), Medium (34.25%), and High (29.81%) — showing risk is fairly evenly distributed.
Marks per Department (Bar Chart)	Compares Marks and an aggregated measure across departments, with EEE and ECE showing the highest combined marks.
Marks per Family Income (Donut Chart)	Shows the share of marks by family income level — Low (40.04%), High (30.70%), Medium (29.25%) — indicating students from lower-income families make up the largest share of the marks recorded.

8. Key Insights and Findings

- The MECH and EEE departments recorded the highest overall attendance, while CSE recorded comparatively lower attendance, suggesting department-level engagement differences.
- Dropout risk is fairly evenly spread across High, Medium, and Low categories, with no single risk group overwhelmingly dominant — indicating that dropout risk is influenced by multiple combined factors rather than one department or group alone.
- EEE and ECE departments show the strongest academic performance in terms of total marks, while CSE trails behind despite reasonable attendance.
- Students from Low family-income backgrounds account for the largest share (40.04%) of recorded marks, highlighting the importance of continued support and resources for economically disadvantaged students.

- Overall, the analysis suggests that a combination of department, attendance, and family income together shape student performance and dropout risk — reinforcing the need for early, data-driven identification of at-risk students.

9. Recommendations

- Institutions should set up automated attendance-tracking dashboards to flag students falling below a minimum attendance threshold.
- Targeted academic support programs should be introduced for departments and income groups showing lower performance trends.
- Students identified as “High” dropout risk should be proactively counselled and monitored each semester.
- Financial-aid and scholarship programs should continue to prioritize students from low-income families to help close the performance gap.
- SAC dashboards should be reviewed periodically (e.g., monthly) so intervention decisions are based on the most current data.

10. Conclusion

This project successfully demonstrates how SAP Analytics Cloud can be used to analyze real-world educational data and identify key challenges affecting student performance and retention. By building a structured dataset and an interactive story dashboard, patterns relating to attendance, department performance, dropout risk, and family income were clearly visualized. These insights can help educational institutions make informed, proactive decisions to support at-risk students and improve overall academic outcomes.

11. References

- SAP Analytics Cloud Trial — <https://www.sap.com/products/data-cloud/cloud-analytics/trial.html>
- SAP Analytics Cloud Help Documentation (in-app)